

Mamun Or Rashid

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Research Interests

Power Systems | Renewable Energy Systems | Microgrids | Smart Grids | Distributed Control | Frequency Control | Cyber-Physical Systems | Cybersecurity of Power Systems | Energy Management | Data-Efficient Machine Learning | Deep Learning | Transfer Learning | Physics-informed AI

Education

Rajshahi University of Engineering & Technology

Rajshahi, Bangladesh

B.SC. IN MECHATRONICS ENGINEERING

Expected July 2026

- **CGPA:** 3.89/4.00 (2nd of 58 students)
- **Thesis:** “A Data-Driven Distributed Framework for Cyber-Resilient Frequency Support of Isolated Network Microgrids”
- **Relevant Coursework:** Artificial Intelligence, Embedded Systems, Robotics, Power Electronics and Drives, Control Systems, Digital Signal Processing and Machine Vision [Detailed Course Curriculum]

Publications

PEER-REVIEWED JOURNAL **M. O. Rashid**, M. O. Islam, S. Billah, S. Tasnim, and S. K. Sarker. (2026). Domain-guided Intelligent Distributed Control for Enhancing Frequency Resilience in Multi-Microgrids under Limited and Noisy Data. *Energy Conversion and Management: X*, 30, 101896. <https://doi.org/10.1016/j.ecmx.2026.101896>

PEER-REVIEWED CONFERENCE S. Mondal, **M. O. Rashid**, S. P. Biswas, and M. R. Islam. (2025). An Optimized Switching Scheme to Reduce the Leakage Current and Power Loss of Transformerless Inverter. In *2025 IEEE International Conference on Mechatronics (ICM)*, Wollongong, Australia, pp. 1–6. <https://doi.org/10.1109/ICM62621.2025.10934771>

MANUSCRIPTS IN PREPARATION **M. O. Rashid**, T. Sultana, M. O. Islam and S. K. Sarker. Enhancing Cyber-resilient Frequency Control of Isolated Microgrids Under Composite Noise: A Collaborative Learning-Based Distributed Framework

Research Experience

Rajshahi University of Engineering & Technology

Rajshahi, Bangladesh

UNDERGRADUATE THESIS RESEARCHER

Ongoing

- **Thesis:** “A Data-Driven Distributed Framework for Cyber-Resilient Frequency Support of Isolated Network Microgrids”
- Developed distributed attack detection framework validated on a two-area isolated microgrid with multiple cyberattack scenarios using MATLAB/Simulink.
- Ongoing work includes MATLAB/Simulink simulation, manuscript preparation, and thesis writing.
- Resulting work published as a first-author Q1 journal article (see Publications); a related manuscript on cyber-resilient frequency control is in preparation

Rajshahi University of Engineering & Technology

Rajshahi, Bangladesh

RESEARCH COLLABORATOR – POWER ELECTRONICS

2025

- Contributed to research on an optimized switching scheme for transformerless inverters, reducing leakage current and power loss in grid-connected photovoltaic systems
- Resulting work published at the 2025 IEEE International Conference on Mechatronics (ICM), Wollongong, Australia (see Publications)

Projects

Brake System Plausibility Device (BSPD) (Formula SAE EV Safety System)

🐙 GitHub

- Designed a standalone analog BSPD compliant with Formula SAE EV regulations, integrating signal conditioning, fault detection, timing logic, and shutdown control on a custom KiCad PCB.

Low EMI Synchronous Step-Down Converter (LMQ61460-Q1)

🐙 GitHub

- Developed a 4-layer Low EMI synchronous step-down converter using the LMQ61460-Q1 for converting 26V to 5V at 6A.

Advance IoT Based Remote Patient Health Monitoring System



- Developed an ESP32-based IoT health monitoring system integrating DHT11 and MAX30100 sensors with real-time remote visualization through the Blynk platform.

Technical Skills

- **Programming:** Python (Advanced), MATLAB (Advanced), C++ (Intermediate)
- **Simulation:** MATLAB/Simulink, LTspice, PSpice, Proteus
- **Machine Learning:** Deep Learning, Transfer Learning, NARX, LSTM
- **Power Systems:** Microgrids, Power Electronics, Frequency Control
- **Embedded Systems:** STM32, ESP32, Arduino
- **PCB Design:** KiCad, Altium
- **Research Tools:** LaTeX, Overleaf, Git, Google Scholar, Zotero

Leadership Experience

2025 - Present **Head of Electrical, Team Crack Platoon** *Rajshahi, Bangladesh*

- Lead a 20-member electrical division across HV, LV, data acquisition, and embedded systems sub-systems, coordinating design reviews, integration milestones, and safety documentation for an international-class EV race car.

2023 - 2025 **Chief of Data Acquisition Systems, Team Crack Platoon** *Rajshahi, Bangladesh*

- Designed the Brake System Plausibility Device (BSPD) from scratch to detect open/short-circuit faults and open the shutdown circuit on failure.
- Contributed to APPS signal validation, LV power distribution, and authored ESF and FMEA documentation for technical inspection.

Honors & Awards

2026 **Student of the Year Award**, Awarded for CGPA 4.00/4.00 across two consecutive semesters; currently ranked 2nd (joint) in the department. *Dept. of MTE, RUET*

2023 - 2026 **Technical Scholarship (4 consecutive years)**, Awarded for outstanding academic performance, academic years 2023, 2024, 2025, and 2026 *RUET*

2025 **18th Overall Ranking**, Formula SAE Japan – EV Class *Worldwide*

2024 **15th Overall Static Result**, Formula Bharat – EV Class *Worldwide*

2023 **MATLAB Onramp Certification**, MathWorks Online Course *Certified by Mathworks*

Professional Activities

Team Crack Platoon *Rajshahi, Bangladesh*

FORMULA SAE JAPAN 2025 & FORMULA BHARAT 2024 *2023 – 2025*

- Designed the Brake System Plausibility Device (BSPD) for EV
- Led HV/LV integration and onboard data-acquisition systems
- Designed PCBs for electrical and data-acquisition subsystems

WALTON Hi-Tech Industries PLC *Gazipur, Bangladesh*

INDUSTRIAL ATTACHMENT *Mar 2023 – Apr 2023*

- Hands-on exposure to SolidWorks-based design, manufacturing processes, industrial robotics, and PLC systems